

/*Write a program in C to detennine whether a number a is prime
or not .

*/

#include <stdio.h>

#include <conio.h>

main()

{

int n, i, flag=0;

printf("Enter a positive number : ");

scanf("%d",&n);

for(i=2;i<=n/2;++i)

{

if(n%i==0)

{

flag=1;

break;

}

}

if (flag==0)

printf("%d is a prime number.",n);

else

printf("%d is not a prime number.",n);

getch();

}

/*Write a program that will obtain the length and width of a rectangle from the user and compute its area and perimeter.

*/

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    float length, width, area, perimeter;
```

```
    printf("Enter the length and width of rectangle:");
```

```
    scanf("%f%f",&length,&width);
```

```
    area=length*width;
```

```
    perimeter=2*length+2*width;
```

```
    printf("Result: %d",perimeter);
```

```
    getch();
```

```
}
```

/*page-76

An equation of the form

$$ax^2 + bx + c = 0$$

is known as the quadratic equation. The values of x that satisfy the equation are known as the roots of the equation.

A quadratic equation has two roots which are given by the following two formulae.

$$\text{root1} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$\text{root2} = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

The program requests the user to input the values of a , b and c and output roots 1 and root2. */

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#include<math.h>
```

```
main()
```

```
{
```

```
    float a, b, c, discriminant, root1, root2;
```

```
    printf ("Input values of a, b, and c\n ");
```

```
    scanf ("%f %f %f", &a, &b, &c);
```

```
    discriminant=b*b-4*a*c;
```

```
    if(discriminant<0)
```

```
        printf ("\n \n Roots are Imaginary\n");
```

```
    else
```

```
    {
```

```
        root1=(-b+sqrt(discriminant) )/(2.0*a);
```

```
        root2=(-b-sqrt(discriminant))/(2.0*a);
```

```
        printf("\n \n Root1=%5.2f\n\n Root2=%5.2f\n",root1,root2);
```

```
    }
```

```
    getch();
```

```
}
```

```
/* C Program to calculate the factorial of an integer .
```

```
*/
```

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
long i,n,fac=1;
```

```
printf("Enter value of n:");
```

```
scanf("%ld",&n);
```

```
for(i=n;i>=1;--i)
```

```
fac*=i;
```

```
printf("\nFactorial of %ld is %ld",n,fac);
```

```
getch(); //to stop the screen
```

```
}
```

```
/* C program to check whether a number is prime or not. */
```

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, i, flag=0;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d",&n);
```

```
    for(i=2;i<=n/2;++i)
```

```
    {
```

```
        if(n%i==0)
```

```
        {
```

```
            flag=1;
```

```
            break;
```

```
        }
```

```
    }
```

```
    if (flag==0)
```

```
        printf("%d is a prime number.",n);
```

```
    else
```

```
        printf("%d is not a prime number.",n);
```

```
    return 0;
```

```
}
```

/* Consider the following Conditional statement:-

```
if(a>b)
```

```
    x=a;
```

```
else
```

```
    x=b;
```

Write the equivalent code segment using a ternary operator pair "?:"

Ans:- `a>b?x=a:x=b;`

```
*/
```

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    int a,b,x;
```

```
    a>b?x=a:x=b;
```

```
    if(a>b)
```

```
        x=a;
```

```
    else
```

```
        x=b;
```

```
    getch();
```

```
}
```

Enter the Radius of circle and Find out its Area and Perimeter.

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
float radius , area;
```

```
float PI=3.142 ;
```

```
printf(" Enter the radius\n");
```

```
scanf("%f" , &radius);
```

```
area= PI * radius * radius ;
```

```
printf(" Area of circle=%f\n " , area);
```

```
printf(" Perimeter=%f\n" ,pmeter);
```

```
getch();
```

```
}
```

Fibonacci Series:

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
main()
```

```
{
```

```
    int a,i,F=0, S=1,T;
```

```
    printf("the limit");
```

```
    scanf("%d",&a);
```

```
    for(i=0;i<=a;i++)
```

```
    {
```

```
        printf("%d \n",F);
```

```
        T=F+S;
```

```
        F=S;
```

```
        S=T;
```

```
    }
```

```
    getch();
```

```
}
```


/*Write a program to compute the sum of the digits of a given integer number.

*/

#include<stdio.h>

#include<conio.h>

main()

{

int num,sum=0,r;

printf("Enter a number:");

scanf("%d",&num);

for(num!=0;num=num/10;num++)

{

r=num%10;

sum=sum+r;

printf("sum of digits of number:%d",sum);

}

getch();

}

/*Write a cprogram to inter change the values
of two given integer.

*/

#include<stdio.h>

#include<conio.h>

main()

{

int x,y,t;

printf("Enter the value of x&y \n:");

scanf("%d%d",&x,&y);

printf("Before Swapping \n x=%d\n y=%d\n",x,y);

t=x;

x=y;

y=t;

printf("After Swapping \n x=%d\n y=%d\n",x,y);

getch();

}

```
/*Write a C program to computer and print a multiplication table
```

```
for numbers 1 to 5
```

```
*/
```

```
#include <stdio.h>
```

```
#include <conio.h>
```

```
main()
```

```
{
```

```
    int n,i;
```

```
    for(n=1;n<=5;n=n+1){
```

```
        for(i=1;i<=10;i=i+1){
```

```
            printf("%d X %d=%d\n",n,i,n*i);
```

```
        }
```

```
    }
```

```
    getch();
```

```
}
```

/*Write a C program to determine whether a number is odd or
Even and print the message

Number Is Even

or

Number Is Odd */

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number is here:");
```

```
    scanf("%d",&n);
```

```
    if(n%2==0)
```

```
    {
```

```
        printf("Number is Even");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Number is Odd");
```

```
    }
```

```
    getch();
```

```
}
```

```
//Program to Calculate perimeter and area of circle
```

```
/*Program to find perimeter and area of circle*/
```

```
#include <stdio.h>
```

```
#include <math.h>
```

```
#include <conio.h>
```

```
main()
```

```
{
```

```
float r,p,a;
```

```
printf("Enter the radius of Circle:");
```

```
scanf("%f",&r);
```

```
p=2*3.14*r;
```

```
a=3.14*r*r;
```

```
printf("The perimeter of circle is:%f\n",p);
```

```
printf("The area of circle is:%f\n",a);
```

```
getch();
```

```
}
```

Recursive function for factorial in c:

```
#include<stdio.h>

int fact(int);

int main(){

    int num,f;

    printf("\nEnter a number: ");

    scanf("%d",&num);

    f=fact(num);

    printf("\nFactorial of %d is: %d",num,f);

    return 0;

}


int fact(int n){

    if(n==1)

        return 1;

    else

        return(n*fact(n-1));

}
```

/* write a program that will convert the given temperature
in fahrenheit in C celsius using the following formula.

$C = (F - 32) / 1.8$

*/

#include<stdio.h>

#include<conio.h>

main()

{

float c,f;

printf("Enter the temperature in Fahrenheit:");

scanf("%f",&f);

c=((f-32)/1.8);

printf("The temperature in celsius is =%f",c);

getch();

}

```
/*Write a program in C to Calculate
the area and circumference of a circle.
*/

#include<stdio.h>
#include<conio.h>
#include<math.h>
main()
{
    int r;
    float area, circumference;
    printf("Enter the value of radius is here:\n");
    scanf("%d",&r);
    circumference=2*3.1416*r;
    printf("The total result=%f, ",circumference);
    getch();
}
```


/*Write a c program to compute the grade of a subject

according the following rules:

subject's mark Grade

>=80+ A

70-79 B

60-69 C

50-59 D

40-49 E

<40 F

*/

#include<stdio.h>

#include<conio.h>

main()

{

int m1, m2, m3, m4, m5,per;

printf("Enter the 5 subject number:");

scanf("%d %d %d %d %d",&m1,&m2,&m3,&m4,&m5);

per=(m1+m2+m3+m4+m5)/5;

if(per>=80&&per<=100)

{

printf("Grade=A");

}

else if (per>=70&&per<=79)

{

printf("Grade=B");

}

else if (per>=60&&per<=69)

{

printf("Grade=C");

```
}  
else if (per>=50&&per<=59)  
{  
    printf("Grade=D");  
}  
else if (per>=40&&per<=49)  
{  
    printf("Grade=E");  
}  
else  
{  
    printf("Grade=F");  
}  
getch();  
return 0;  
}
```

```
/*PROGRAM TO FIND SUM OF ALL INTEGER WHICH IS > 100 AND< 200
```

```
AND WHICH IS DIVISIBLE BY 7*/
```

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    int i,sum = 0;
```

```
    //scanf("%d",&i);
```

```
    for(i = 0;i <= 300 ; i++)
```

```
    {
```

```
        if (i % 9== 0)
```

```
        sum = sum + i;
```

```
    }
```

```
    printf("\n Sum of all no between 0 and 300 ");
```

```
    printf("which is divisible by 9 is :: %d",sum);
```

```
    getch();
```

```
}
```

```
/*
```

```
*****
```

```
OUTPUT
```

```
*****
```

```
Sum of all no between 100 and 200 which is divisible by 7 is :: 2107
```

```
*/
```

Programming solutions E BALAGURUSAMY

Chapter-1

Problem-1.1

Write a program that will print your mailing address in the following form:

First Line: Name

Second Line : Door No, Street

Third Line: City, Pin code

Solution:

```
#include <stdio.h>
void main()
{
printf("Sanjida Afrin Moul\n");
printf("35 no Hamidra Dash road\n");
printf("Dhaka\n");
}
```

Problem -1.2

Modify the above program to provide border line to the address.

```
#include <stdio.h>
#include <stdlib.h>
void main()
{
printf("Sanjida Afrin Moul\n");
printf("|-----|\n");
printf("|35 no Hamidra Dash road |\n");
printf("|Dhaka |\n");
printf("|-----|");
}
```

Problem-1.3

Write a program using one print statement to print the pattern as shown below

```
#include <stdio.h>
#include <stdlib.h>
void main()
{
printf("*\n**\n***\n****\n");
}
```

Problem-1.4

Given the radius of a circle, write a program to compute and display its area. Use a symbolic constant to define PI value and assume a suitable value for radius.

Solution:

```
#include <stdio.h>
#include <stdlib.h>
#define PI 3.1416
void main()
{
    int radius;
    float area;
    scanf("%d",&radius);
    area=PI*(radius*radius);
    printf("The area of the circle is %f",area);
}
```

Problem1.5

Write a program to output the following multiplication table

5x1=5

5x2=10

.....

.....

5x10=50

Solution:

```
#include <stdio.h>
#include <stdlib.h>
void main()
{
    int num1=5,num2,result;
    for(num2=1;num2<=10;num2++)
    {
        result=num1*num2;
        printf("%d x %d=%d\n",num1,num2,result);
    }
}
```

Problem1.6

Given two integers 20 and 10, write a program that uses a function add() to add these two numbers and sub() to find the difference of these two numbers and then display the sum in the following form-

20+10=30

20-10=10

Solution:

```
#include <stdio.h>
#include <stdlib.h>
```

```

int add(int x, int y);
int sub(int x, int y);
main()
{
int sum,dif;
sum=add(10,20);
printf("20+10=%d\n",sum);
dif=sub(20,10);
printf("20-10=%d",dif);
}
nt add(int x, int y) //function definition
{
int d;
d=x+y;
return(d);
}
nt sub(int x, int y) //function definition
{
int d;
d=x-y;
return(d);
}

```

Problem 1.7

Given the values of three variables a, b and c, write a program to compute and display the value of x, where $x=a/(b-c)$.

Execute your proram for the following values:

(a) a=250, b=85, c=25

(b) a=300, b=70, c=70

Solution:

```

#include <stdio.h>
#include <stdlib.h>
main()
{
float x;
int a, b, c;
printf("Value of a\n");
scanf("%d",&a);
printf("Value of b\n");
scanf("%d",&b);
printf("Value of c\n");
scanf("%d",&c);
x=a/(b-c);
printf("The value of x is %f",x);
}

```

Problem 1.8

Relationship between Celsius and Fahrenheit is governed by the formula $F = (9C/5) + 32$

Write a program to convert temperature

(a) From Celsius to Fahrenheit

(b) From Fahrenheit to Celsius

Solution:

```
#include <stdio.h>
#include <stdlib.h>
main()
{
    float f,c,result;
    char option;
    printf("Enter your option F or C F for celcius to Fahrenheit and C for Fahrenheit to Celcius\n");
    option=getchar();
    if(option=='F' || option=='f')
    {
        printf("Input Celcius\n");
        scanf("%f",&c);
        result=((9*c)/5)+32;
        printf("The Fahrenheit is %f",result);
    }
    else if(option=='C' || option=='c')
    {
        printf("Input Fahrenheit\n");
        scanf("%f",&f);
        result=(5*(32-f))/9;
        printf("The Celcius is %f",result);
    }
    else
    {
        printf("Wrong option chosen");
    }
}
```

Problem 1.9

Area of a triangle is given by the formula $A = \sqrt{S(S-a)(S-b)(S-c)}$

Where a, b and c are sides of the triangle and $2S = a + b + c$. Write a program to compute the area of the triangle given the values a, b and c.

Solution:

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>
main()
{
    int a, b, c;
    float s, result;
    printf("Enter value of a");
```

```

scanf("%d",&a);
printf("Enter value of b");
scanf("%d",&b);
printf("Enter value of c");
scanf("%d",&c);
s=(a+b+c)/2;
result=sqrt(s*(s-a)*(s-b)*(s-c));
printf("Area is %f",result);
}

```

Problem-1.10

Distance between two points (x1,y1) and (x2,y2) is governed by the formula

$$D^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

Write a program to compute D given the coordinates of the points.

Solution:

```

#include <stdio.h>
#include <stdlib.h>
#include <math.h>
main()
{
int x1,x2,y1,y2;
float d;
printf("Insert X1");
scanf("%d",&x1);
printf("Insert X2");
scanf("%d",&x2);
printf("Insert Y1");
scanf("%d",&y1);
printf("Insert Y2");
scanf("%d",&y2);
d=sqrt((x2-x1)*(x2-x1)+(y2-y1)*(y2-y1));
printf("Value of D is %f",d);
}

```

Problem 1.11

A point on the circumference of a circle whose center is (0,0) and (5,6). Write a program to compute perimeter and area of the circle.

```

#include <stdio.h>
#include <math.h>
#define pi 3.14159
void main()
{
float r,x1,x2,y1,y2,A;
x1=0;
x2=0;

```



```

y1=4;
y2=5;
r=sqrt((x1-x2)*(x1-x2)+(y1-y2)*(y1-y2));
A=pi*r*r;
printf("Result=%f",A);
}

```

Problem-1.12

The line joining the points (2,2) and(5,6) which lie on the circumference of a circle is the diameter of the circle. Write a program to compute the are of the circle.

```

#include<stdio.h>
#include<math.h>
#define pi 3.14159
void main()
{
float D,r,x1,x2,y1,y2,A;
x1=2;
x2=2;
y1=5;
y2=6;
D=sqrt((x1-x2)*(x1-x2)+(y1-y2)*(y1-y2));
r=D/2;
A=pi*r*r;
printf("Result=%f",A);
}

```

Problem 1.14

Write a program to make the equation of a line in form $ax+by=c$
For $a=5$, $b=8$, $c=18$

Solution:

```

include<stdio.h>
#include<conio.h>
void main()
{ int a,b,c;
clrscr();
a=5;
b=8;
c=18;
printf("%dx+%dy=%d",a,b,c);
getch();
}

```

/* page-53

program-3.1

The program in fig 3.1 shows the use of integer arithmetic to convert a given number of days into months and day.*/

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    int months, days;
```

```
    printf("Enter days \n ");
```

```
    scanf("%d",&days);
```

```
    months=days/30;
```

```
    days=days%30;
```

```
    printf("Month=%d Days=%d",months,days);
```

```
    getch();
```

```
}
```

/* page-58

program-3.2

The program in fig 3.2 prints a sequence of squares of numbers.

Note the use of the shorthand operator *=

*/

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#define N 100
```

```
#define A 2
```

```
main()
```

```
{
```

```
    int a;
```

```
    a=A;
```

```
    while(a<N)
```

```
    {
```

```
        printf("%d\n",a);
```

```
        a*=a;
```

```
    }
```

```
    getch();
```

```
}
```

/* page-70

program-3.7

The program shows a program using a cast to evaluate the equation $\sum_{i=1}^n (1/i)$ */

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    float sum;
```

```
    int n;
```

```
    sum=0;
```

```
    for(n=1;n<=10;++n)
```

```
    {
```

```
        sum=sum+1/(float)n;
```

```
        printf("%2d %6.4f\n",n,sum);
```

```
    }
```

```
    getch();
```

```
}
```

/* page-74

program-

A computer manufacturing company has the
following monthly compensation policy to their sales-persons:

Minimum base salary:1500.00

Bonus for every computer sold:200.00

commission on the total monthly sales:2per cent

*/

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#define BASE_SALARY 1500
```

```
#define BONUS_RATE 200
```

```
#define COMMISSION 0.02
```

```
main()
```

```
{
```

```
    int quantity;
```

```
        float gross_salary,price;
```

```
        float bonus,commission;
```

```
        printf("Input number sold and price\n");
```

```
        scanf("%d %f",&quantity,&price);
```

```
        bonus=BONUS_RATE*quantity;
```

```
        commission=COMMISSION*quantity*price;
```

```
        gross_salary=BASE_SALARY+bonus+commission;
```

```
        printf("\n");
```

```
        printf("Bonus=%f\n",bonus);
```

```
        printf("Commission=%f\n",commission);
```

```
        printf("Gross salary=%f",gross_salary);
```

```
        getch();
```

```
}
```

/*page-76

An equation of the form

$$ax^2 + bx + c = 0$$

is known as the quadratic equation. The values of x that satisfy the equation are known as the roots of the equation.

A quadratic equation has two roots which are given by the following two formulae.

$$\text{root1} = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

$$\text{root2} = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

The program requests the user to input the values of a , b and c and output roots 1 and root2. */

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#include<math.h>
```

```
main()
```

```
{
```

```
    float a, b, c, discriminant, root1, root2;
```

```
    printf ("Input values of a, b, and c\n ");
```

```
    scanf ("%f %f %f", &a, &b, &c);
```

```
    discriminant=b*b-4*a*c;
```

```
    if(discriminant<0)
```

```
        printf ("\n \n Roots are Imaginary\n");
```

```
    else
```

```
    {
```

```
        root1=(-b+sqrt(discriminant) )/(2.0*a);
```

```
        root2=(-b-sqrt(discriminant))/(2.0*a);
```

```
        printf("\n \n Root1=%5.2f\n\n Root2=%5.2f\n",root1,root2);
```

```
    }
```

```
    getch();
```

```
}
```

/*page-87

A program that reads a character from keyboard and then prints in reverse case in given in fig.4.3 . that is, if the input is upper case, the output will be lower case and vice versa. */

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#include<ctype.h>
```

```
main()
```

```
{
```

```
    char alphabet;
```

```
    printf("Enter an alphabet ");
```

```
    putchar('\n'); /*move to next line*/
```

```
    alphabet=getchar();
```

```
    if(islower(alphabet))
```

```
        putchar(toupper(alphabet));/*Reverse and display*/
```

```
    else
```

```
        putchar(tolower(alphabet));/*Reverse and display*/
```

```
        getch();
```

```
}
```

/*page-115

The program counts the number of boys whose weight is less than 50kg and height is greater than 170cm.

*/

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    int count, i;
```

```
    float weight, height;
```

```
    count=0;
```

```
    printf("Enter weight and height for 10 boys\n ");
```

```
    for(i=1;i<=10;i++);
```

```
    {
```

```
        scanf("%f %f",&weight, &height);
```

```
        if(weight<50 && height>170)
```

```
            count=count+1;
```

```
    }
```

```
    printf("Number of boys with weight < 50kg\n");
```

```
    printf("and height > 170 cm = %d\n",count);
```

```
    getch();
```

```
}
```




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